

1. Multiple Catch

```
public class Main {  
  
    public static void main(String[] args) {  
  
        try {  
  
            int a = 10 / 0;  
  
            int[] arr = new int[5];  
            arr[10] = 100;  
  
        } catch (ArithmeticException e) {  
  
            System.out.println("ArithmeticException: Cannot divide by zero.");  
  
        } catch (ArrayIndexOutOfBoundsException e) {  
  
            System.out.println("ArrayIndexOutOfBoundsException: Invalid Array Index.");  
  
        } catch (Exception e) {  
  
            System.out.println("Exception Handled.");  
        }  
    }  
}
```

Output

```
ArithmeticException: Cannot divide by zero.
```

```
=== Code Execution Successful ===
```

2. StringIndexOutOfBoundsException

```
public class Main {  
  
    public static void main(String[] args) {  
  
        try {
```

```

        String str = "JAVA";

        System.out.println(str.charAt(10));

    } catch (StringIndexOutOfBoundsException e) {

        System.out.println("StringIndexOutOfBoundsException: Invalid String Index.");
    }
}
}

```

Output

```
StringIndexOutOfBoundsException: Invalid String Index.
```

```
=== Code Execution Successful ===
```

3. finally

```

public class Main {

    public static void main(String[] args) {

        try{

            int result = 20 / 2;

            System.out.println("Result = " + result);

        } catch (ArithmeticException e) {

            System.out.println("Exception Occurred.");

        } finally{

            System.out.println("Finally Block Executed.");
        }
    }
}

```

Output

```
Result = 10  
Finally Block Executed.  
  
=== Code Execution Successful ===
```

4. throw (Check Person is Eligible to Vote)

```
import java.util.Scanner;  
  
public class Main {  
  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        try {  
  
            System.out.print("Enter Age: ");  
            int age = sc.nextInt();  
  
            if (age < 18) {  
                throw new ArithmeticException("Not Eligible to Vote");  
            }  
  
            System.out.println("Eligible to Vote");  
  
        } catch (ArithmeticException e) {  
  
            System.out.println(e.getMessage());  
        }  
  
        sc.close();  
    }  
}
```

Output

```
Enter Age: 20  
Eligible to Vote
```

```
=== Code Execution Successful ===
```

5. throws

```
import java.io.*;  
  
public class Main {  
  
    static void checkFile() throws IOException {  
  
        FileReader file = new FileReader("test.txt");  
  
        System.out.println("File Opened Successfully.");  
  
        file.close();  
    }  
  
    public static void main(String[] args) {  
  
        try {  
  
            checkFile();  
  
        } catch (IOException e) {  
  
            System.out.println("IOException: File Not Found.");  
        }  
    }  
}
```

Output

```
IOException: File Not Found.
```

```
=== Code Execution Successful ===
```